Bruno Gehlen F. da Silva

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"We are what we repeatedly do. Excellence, then, is not an act, but a habit." – Aristotle

Personal Profile

Hi! I hold a bachelor's degree in Physics from USP, where I cultivated a deep fascination for theoretical frameworks—particularly particle physics and the profound mathematical structures that underpin it. This passion has since guided me toward pure mathematics, where I am currently pursuing a master's degree focused on advancing solutions to the Yamabe Problem. My research leverages tools from differential geometry and functional analysis to explore the existence and classification of conformal metrics with constant scalar curvature on Riemannian manifolds—a challenge that bridges geometric analysis, elliptic PDEs, and theoretical physics.

The interplay between abstract mathematics and physical intuition drives my work; I am particularly drawn to how geometric insights can transcend traditional boundaries, offering fresh perspectives on fundamental questions beyond the Standard Model. Alongside my academic pursuits, I nurture a creative streak in UI/UX design and front-end development. Crafting intuitive, visually compelling interfaces allows me to merge analytical rigor with artistic expression, and I continually seek opportunities to expand my technical toolkit in this evolving field.

Whether navigating the intricacies of conformal transformations or coding responsive web layouts, I thrive on turning complex ideas into elegant, functional solutions.

Education

Institute of Mathematics and Statistics at the University of São Paulo — *Sao Paulo-SP, Brazil*
M.Sc in Mathematics — *April 2025 - Currently*

Institute of Physics at the University of São Paulo — *Sao Paulo-SP, Brazil*
B.Sc in Physics — *2021 - 2024*

Sartre | Escola SEB — *Lauro de Freitas-BA, Brazil*
High School — *2018 - 2020*

Work Experience

Master's Project - Institute of Mathematics and Statistics IME-USP — *São Paulo, Brazil*
Classification of solutions to the singular Yamabe problem and their relation to gravitational instantons — *Feb 2025 - Currently*

- This project proposes an in -depth study on the classification of positive solutions of the singular Yamabe problem in compact Riemannian varieties and their interpretation as gravitational instant in gravity theories governed by scalar curvature. The proposal connects classic analytical and geometric techniques with modern tools, including the positive mass theorem and extensions for fourth order theories through Paneitz-Branson operators. It is intended to understand the role of these solutions in the context of gravitational collapse and formation of singularities, with potential application to mathematical physics and geometric theory of fields.

Scientific Initiation - Institute of Theoretical Physics IFT UNESP — *São Paulo, Brazil*
Mathematica Demonstrations in an introduction to Quantum Field Theory and Particle Physics — *Jun 2024 - Dec 2024*

- It is of paramount importance to expose undergraduate students to cutting-edge physics topics as soon as they have the capacity to grasp them. In this project, our aim is to enhance candidates' education by delving into introductory texts on Particle Physics and Field Theory. In addition to the usual approach of reading books and presentations, we will complement it with the creation of Mathematica programs that will perform relevant calculations for this study. These programs will not only facilitate simulations and demonstrations, offering candidates a deeper understanding of the subject, but will also be made available to other students or the general public.

Scientific Initiation - Institute of Physics IF-USP — *São Paulo, Brazil*
Study of Hadron Production from Thermal Models in Collisions between Relativistic Heavy Ions — *Aug 2023 - Jun 2024*

- This project aims to study the production of hadrons composed of heavy quarks using thermal models in order to assess whether there is evidence of thermalization of these particles in the medium formed in collisions between relativistic heavy ions. Studies of hadron production using a thermodynamic approach are of interest on several fronts, especially with regard to the hadronization of the medium formed in these collisions. The possible finding that charm and beauty quarks exhibit similar behavior to lighter quarks during hadronization may reveal an intricate mechanism of thermalization of these particles with the medium formed. In addition to the scientific interest of this investigation, the didactic nature of this approach should be highlighted, as it is suitable for students beginning the advanced cycle of a bachelor's degree in physics.

Extracurricular

Curso de Verão do IF-USP — *São Paulo-SP, Brazil*
Universidade de São Paulo — *Feb 2023*

IV Escola de Inverno Jayme Tiomno — *São Paulo-SP, Brazil*
Universidade de São Paulo — *Aug 2022*

- Física Além do Modelo Padrão
- Introdução aos Modelos Científicos a Partir da Filosofia da Ciência: Uma Abordagem Pela Física
- Sistemas Hamiltonianos em Espaços Vetoriais Simpléticos
- Uma Introdução À Teoria Quântica de Campos em Espaços-Tempos Curvos

Curso de Verão do IF-USP — *São Paulo-SP, Brazil*
Universidade de São Paulo — *Mar 2022*

III Escola de Inverno Jayme Tiomno — *São Paulo-SP, Brazil*
Universidade de São Paulo — *Aug 2021*

- Introdução aos Fundamentos Filosóficos da Mecânica Quântica
- Introdução à Teoria de Supercordas
- Métodos Algébricos da Física Teórica

Skills

Programming
- Python (specifically science/data packages, as Pandas, NumPy, SciPy, Matplotlib)
- Fronted in general (the triad HTML/CSS/JavaScript), PHP
- Wolfram Mathemtica
- R Language
- Arduino

Frameworks
- Node.js, Webpack, Vite
- React, Next.js
- Jupyter, Google Colab

Database
- Vercel
- AWS Amplify
- Heroku
- Github Pages

Miscellaneous Linux, LaTeX, Microsoft Office, A personal Homelab where I take time to test new technologies
Soft Skills Problem-solving, Efficiency Enthusiast,Documentation, Organized, Optimizer.

Achievements

2019	**Bronze Medal**, Mathematical Kangaroo	*Brazil*
2020	**Gold Medal**, National Science Olympiad (ONC)	*Brazil*
2020	**Bronze Medal**, Brazilian Olympics of Astronomy (OBA)	*Brazil*

Interests

Physics I have always been passionate about it, especially the mathematical formalism. I am currently applying for a master's degree in pure mathematics at Institute of Mathematics and Statistics of USP.
Open source Since 2021, I have been exploring the world of Linux and the open source. Currently using Arch.
Music I enjoy almost all types of music, as long as it's Brazilian. I particularly listen to rap, R&B and jazz.
Art I have always enjoyed drawing since I was a child. Sometimes I sketch something to distract myself.

Languages

Portuguese Native proficiency
English Bilingual proficiency
French Beginner proficiency
Spanish Beginner proficiency